

Accuracy and reliability of self-reported data in interorganizational networks *

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Although network analysis is becoming a more viable and informative methodology for the study of interorganizational behavior, issues involving the accuracy and reliability of self-reported or 'cognitive' network data are still unresolved. Two networks based on information and coordination relationships in each of four major cities are examined. Reliability of responses is assessed as the percent of confirmed relationships between agency respondents and by correlating the extent of agreement in each network. In addition, two possible measurement errors are examined: systematic errors which are network-independent and idiosyncratic errors which are network-dependent. These errors are assessed through correlational analyses of the intensity of an organization's relationships with the number of its relationships that are confirmed in each network. Also systematic errors are accounted for by calculating the proportion of confirmed relationships within the different response categories used to judge the strength of relationship between ego organizations in the network. It is concluded that boundary spanning personnel of organizations can provide fairly reliable responses to network generated questions, but that errors in the data tend to be systematic in the way that prior studies of interpersonal cognitive networks report.

1. Introduction

Over the past two decades there has been a rapid increase in the use of network analysis techniques in interorganizational studies (Oliver

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